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Salish Sea Chinook salmon exhibit weaker coherence in early marine survival trends than coastal populations

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Abstract

Identifying factors that influence anadromous Pacific salmon (*Oncorhynchus* spp.) population dynamics is complicated by their diverse life histories and large geographic range. Over the last several decades, Chinook salmon (*O. tshawytscha*) populations from coastal areas and the Salish Sea have exhibited substantial variability in abundance. In some cases, populations within the Salish Sea have experienced persistent declines that have not rebounded. We analyzed a time series of early marine survival from 36 hatchery Chinook salmon populations spanning ocean entry years 1980–2008 to quantify spatial and temporal coherence in survival. Overall, we observed higher inter-population variability in survival for Salish Sea populations than non-Salish Sea populations. Annual survival patterns of Salish Sea populations covaried over smaller spatial scales and exhibited less synchrony among proximate populations relative to non-Salish Sea populations. These results were supported by multivariate autoregressive state space (MARSS) models which predominantly identified region-scale differences in survival trends between northern coastal, southern coastal, Strait of Georgia, and Puget Sound population groupings. Furthermore, Dynamic Factor Analysis (DFA) of regional survival trends showed that survival of southern coastal populations was associated with the North Pacific Gyre Oscillation, a large-scale ocean circulation pattern, whereas survival of Salish Sea populations was not. In summary, this study demonstrates that survival patterns in Chinook salmon are likely determined by a complex hierarchy of processes operating across a broad range in spatial and temporal scales, presenting challenges to the management of mixed-stock fisheries.

KEYWORDS
Chinook salmon, marine survival, multivariate autoregressive state space models, resource management, Salish Sea, spatio-temporal variability, time series

1 | INTRODUCTION

Sound management and conservation of exploited organisms require an understanding of the predominant spatial and temporal scales of variability governing both short- and long-term population dynamics. Identifying the sources of this variability remains a critical challenge to managers tasked with developing, modifying, and implementing resource management strategies (Ludwig, Hilborn, & Walters, 1993). Population dynamics of Pacific salmon (*Oncorhynchus* spp.), specifically life-stage-specific survival or abundance, are particularly difficult

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to accurately project because their life cycle encompasses both freshwater and marine environments (Quinn, 2005), subjecting them to a large suite of factors that influence the overall viability of populations. Correlation in survival and recruitment rates for populations separated by hundreds of kilometers indicate that marine conditions common to a region influence population dynamics similarly (Dorner, Peterman, & Haeseker, 2008; Sharma, Velez-Espino, Wertheimer, Mantua, & Francis, 2013; Malick & Cox, 2016). This relationship occurs at a variety of spatial scales, ranging from those as large as the Northeast Pacific Ocean (Kilduff, Di Lorenzo, Botsford, & Teo, 2015) to as small as local conditions encountered at the point of marine entry (Greene, Jensen, Pess, Steel, & Beamer, 2005; Sharma et al., 2013).

Here, we examine spatial and temporal complexity in marine survival of Chinook salmon (*O. tshawytscha*) populations in the Northeast Pacific, focusing on the Salish Sea and develop testable hypotheses about the spatial and temporal scales at which specific environmental and biological drivers may influence smolt survival patterns. The Salish Sea is a unique inland marine ecosystem encompassing the interconnected waters of the Strait of Georgia and Puget Sound (Figure 1) and connected to the Pacific Ocean via the Strait of Juan de Fuca and Johnstone Strait. Glaciation of the region carved a network of basins and inlets with spatially-explicit oceanographic properties such as salinity, temperature, and vertical stratification (Moore et al., 2008a). Chinook salmon are native throughout the Salish Sea and have experienced large fluctuations in abundance throughout the last several decades (Good, Waples, & Adams, 2005; Peterman & Pyper, 2000; Preikshot, Beamish, & Neville, 2013). These fluctuations may be linked to a combination of human-caused stressors (e.g., habitat loss, overharvest, and hatchery propagation) and natural environmental drivers in both freshwater and marine environments (Lawson, 1993; Peterson & Schwing, 2003; Ruckelshaus, Levin, Johnson, & Kareiva, 2002; Ward, Anderson, Beechie, Pess, & Ford, 2015). Declines in abundance have led to conservation measures such as protection under the U.S. Endangered Species Act for Puget Sound populations (Ford et al., 2011; Ruckelshaus et al., 2006) and efforts to assess and implement recovery measures for Strait of Georgia populations. Declines in the marine survival of Puget Sound coho salmon (Zimmerman et al., 2015) and Steelhead trout (Kendall, Marston, & Klungle, In press), Strait of Georgia Chinook and coho salmon (Preikshot et al., 2013), and concurrent changes in pelagic community structure (Greene, Kuehne, Rice, Fresh, & Penttila, 2015) raise

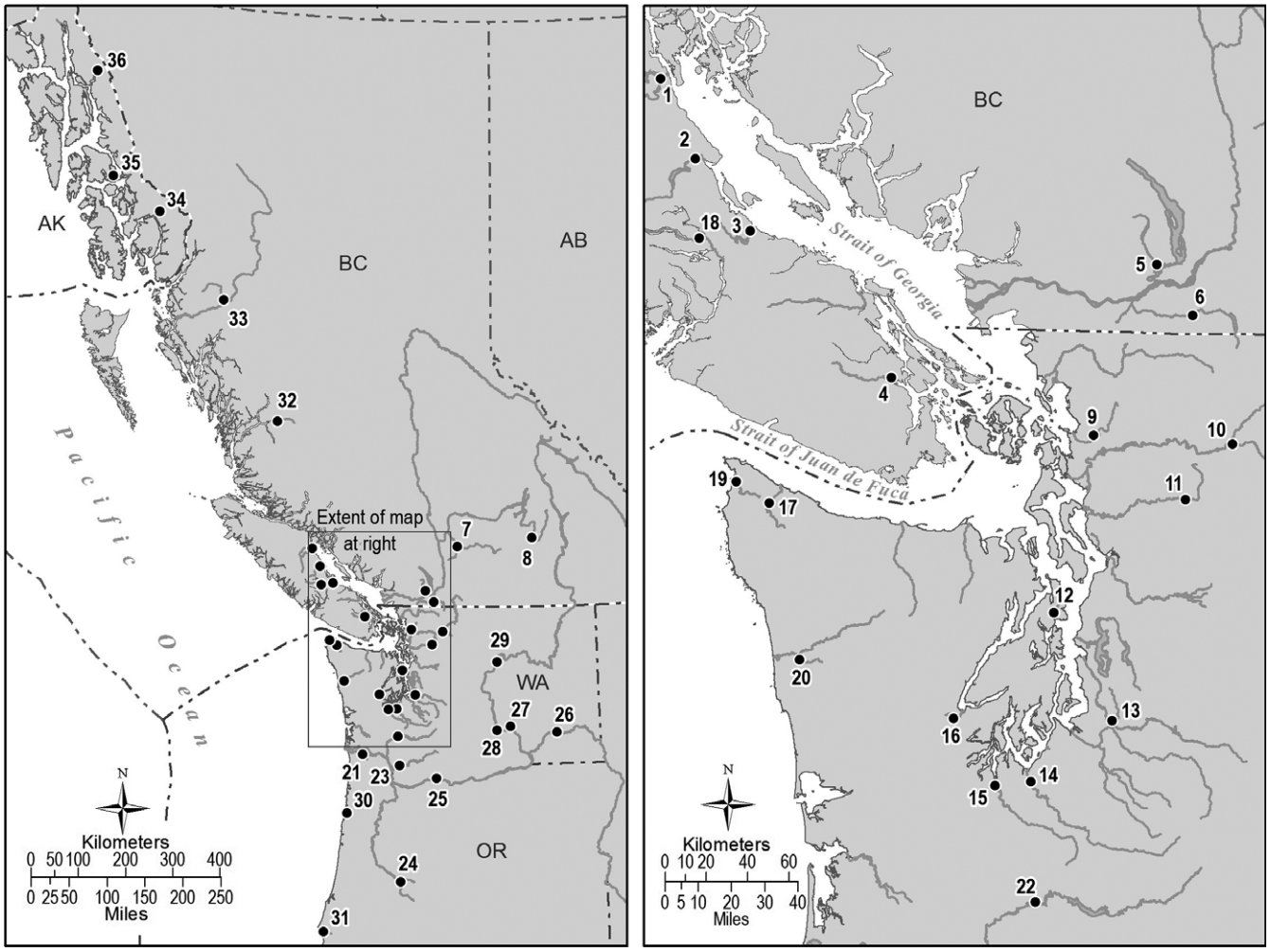


FIGURE 1 Study area showing the geographic extent of Chinook salmon populations included in analyses with the Salish Sea shown in reference to the spatial extent of the study. Numbers correspond to population information provided in Table 2

the possibility that environmental change unique to the Salish Sea ecosystem has contributed to salmon declines. Accordingly, one might expect that marine survival of Salish Sea salmon populations is lower than, and potentially asynchronous with, marine survival of populations outside of the Salish Sea.

Why might Salish Sea populations exhibit greater variation in smolt survival than non-Salish Sea populations? As a semi-enclosed body of water, the Salish Sea may be subject to more rapid changes in oceanographic properties than the open Pacific Ocean into which coastal rivers drain. For example, annual variation in river discharge and local air temperature have a large basin-scale influence on the Puget Sound ecosystem and correlate more strongly with Puget Sound oceanographic properties than larger-scale climate indices such as PDO, ENSO, and the Aleutian Low (Moore et al., 2008a). Furthermore, within Puget Sound, environmental variability tends to increase along a gradient from its outer, more oceanic waters (i.e., Admiralty Inlet) to its more distal bays and inlets (Moore, Mantua, Kellogg, & Newton, 2008b). If the California Current off the coast of North America is more strongly moderated by continental scale climatic processes than Puget Sound, and these climate processes fluctuate on longer time frames than Puget Sound air temperature and river discharge, Puget Sound Chinook salmon may encounter a more variable environment than coastal populations during early marine rearing.

The broad geographic scales across which Chinook salmon co-mingle in the marine environment make many populations susceptible to marine fisheries that occur well outside their region of origin (Weitkamp, 2010). The failure of pre-season forecast models to accurately predict early marine survival, which is critical to the overall survival of populations (Beamish & Mahnken, 2001; Beamish, Mahnken, & Neville, 2004), can lead to significant errors in population-specific abundance forecasts (see Scheuerell & Williams, 2005). Forecast error can lead to over- or under- projections of total allowable mortality in marine fisheries contributing to either unnecessary limitations on fisheries or overexploitation (PSC, 2015). Furthermore, the spatial scale at which forecast abundance indices are calculated may not accurately reflect the spatial scale of variability in early marine survival. This discrepancy can lead to disproportionate fishery impacts on weaker populations. Therefore, a better understanding of the predominant spatial scales across which populations co-vary in survival will help to inform appropriate spatial scales of management and assessment for Chinook salmon.

In this study, we quantified spatial and temporal coherence in hatchery origin Chinook salmon smolt survival. Smolt survival is defined as the period from hatchery release to the end of the first year of ocean residence. We placed emphasis on populations originating from within the Salish Sea due not only to their cultural and economic importance to local communities but also their contribution to Northeast Pacific mixed-stock commercial troll and sport fisheries. Because Salish Sea Chinook salmon are harvested extensively in marine mixed-stock fisheries, we used coded wire tag (CWT) data to estimate stock-specific harvest and escapement and calculate smolt survival for 36 hatchery Chinook salmon populations distributed across a broad geographic range, extending from coastal Southeast Alaska to Oregon and within the Salish Sea (Figure 1). However, the quality of smolt survival estimates derived from CWT recovery data can vary substantially by population and year due to poor fishery and escapement sampling coverage (PSCCWTWG, 2008). To address this challenge, we utilized multivariate state space models to assess data support for hypotheses regarding the predominant spatial and temporal scales governing variability in Chinook salmon smolt survival rates (see Table 1 for spatial scale definitions). Recently, these models have received considerable attention regarding their usefulness to evaluate noisy ecological time series (Ohlberger, Scheuerell, & Schindler, 2016; Ward et al., 2010; Zuur, Tuck, & Bailey, 2003;), particularly because of their ability to address problems with missing data and effectively partition the total variance present in a data time series into signal and noise components (Dennis, Ponciano, Lele, Taper, & Staples, 2006), with the latter being important for reducing bias in estimates of survival trends.

This study addresses the following questions: (i) is smolt survival of Chinook salmon populations within the Salish Sea similar to smolt survival of populations outside the Salish Sea (i.e., region scale, Table 1)? (ii) do the two basins within the Salish Sea (Strait of Georgia and Puget Sound) exhibit similar trends in survival? Results from this study will be used to inform the appropriate spatial scales for future work to identify ecosystem indicators for improving the accuracy and precision of stock assessment and forecasting methods necessary for effective management and conservation of Salish Sea Chinook salmon.

2 | MATERIALS AND METHODS

2.1 | Study area and CWT dataset

Total releases and recoveries of CWT Chinook salmon were compiled from three geographic areas: Northern Coastal (Southeast Alaska [SEAK] and Northern British Columbia [NBC]), Salish Sea (Strait of Georgia [SOG], Puget Sound [PS], and Strait of Juan de

TABLE 1 Description of spatial scale terminology used to describe geographic coherence in Chinook salmon smolt survival

Scale	Description
Continental	Entire west coast of North America
Region	Broad regions of western North America: Northern Coastal, Salish Sea, Southern Coastal
Basin	Basins within the Salish Sea: Puget Sound, Strait of Georgia, and Strait of Juan de Fuca States or provinces within the northern region: Southeast Alaska, British Columbia Southern region: Columbia River or coastal
Sub-basin	Areas within Puget Sound (Whidbey, Central, South, Hood Canal) and the Strait of Georgia (Northern, Central, Southern) defined by shared oceanographic attributes
Population	Individual populations in the analysis

Fuca [JDF]), and Southern Coastal (West Coast Vancouver Island, Washington, and Oregon Coast). At least some oceanographic properties in the Strait of Juan de Fuca are transitional between the Salish Sea and Pacific Ocean (Johannessen, Masson, & Macdonald, 2006; Masson & Peña, 2009), so we tested this area as both within and separate from the Salish Sea with MARSS models. Populations were selected for inclusion in analyses based on data quality, time series length (minimum 20 years), and geographic coverage within and outside the Salish Sea, resulting in a list of 36 populations with survival data over ocean entry years (OEY) 1980–2008 (Table 2; Figure 1). This dataset represented the dominant life history types locally observed in wild Chinook salmon populations: ocean entry ages mimicked by two release strategies (subyearling, yearling) and two run-timing groups (spring, summer-fall). CWT recovery data were downloaded from the coast-wide Regional Mark Information System database (http://www.rmpc.org, accessed 6 June 2015).

2.2 | Calculation of smolt to age-2 or 3 survival of Chinook salmon

Release and recovery data from CWT fish were used to estimate population- and brood-specific smolt survival of Chinook salmon using backwards cohort reconstruction. The survival metric encompasses all sources of post-release mortality of CWT fish to age-2 for Chinook salmon released as subyearlings and to age-3 for Chinook salmon released as yearlings. Smolt survival was calculated as follows:

$$\hat{S}_{i,\mathrm{BY},a'} = \frac{\hat{N}_{i,\mathrm{BY},a'}}{R_{i,\mathrm{BY}}} \quad (1)$$

where $\hat{N}_{i,\mathrm{BY},a'}$ is the estimated cohort size of population i, age a' ($a' = 2$ and 3 for subyearling and yearling releases, respectively), and $R_{i,\mathrm{BY}}$ is the number of fish released from population i, brood year BY. Cohort sizes were computed using virtual population analysis (Coronado & Hilborn, 1998) by reconstructing specific cohorts recursively from the oldest age (age-5 for subyearling releases and age-6 for yearling releases) to the youngest age (i.e., a') based on the estimated numbers of CWT fish recovered from population i, brood year BY, at age a in pre-terminal (i.e., ocean) fisheries ($\hat{P}_{i,\mathrm{BY},a}$), terminal (i.e., freshwater) fisheries ($\hat{T}_{i,\mathrm{BY},a}$), and escapement ($\hat{E}_{i,\mathrm{BY},a}$):

$$\hat{N}_{i,\mathrm{BY},a'} = \frac{\hat{P}_{i,\mathrm{BY},a} + \hat{T}_{i,\mathrm{BY},a} + \hat{E}_{i,\mathrm{BY},a} + \hat{N}_{i,\mathrm{BY},a+1}}{1 - M_a} \quad (2)$$

where M_a is the natural mortality occurring in each age before fishing mortality, assumed to be 40% for age-2, 30% for age-3, 20% for age-4, and 10% for age-5 and older Chinook salmon (PSC, 2015; Sharma et al., 2013).

2.3 | Statistical analyses

We used multiple analytical methods to examine the spatial and temporal coherence in salmon smolt survival patterns both with and without prior assumptions of survival pattern groupings according to geography, life history type, and release strategy. Because the focus of this study was to determine the degree of similarity among trends rather than absolute magnitudes, survival time series were logit-transformed and centered to a mean of 0 before model fitting. All analyses were conducted using the R statistical computing language (R Core Team, 2017).

To examine the relationship between geographic proximity and correlation in survival between Chinook salmon populations, we fit an exponential decay model of the form:

$$\rho_d = \rho_0 e^{-d/\nu} \quad (3)$$

where ρ_d is the Pearson's correlation coefficient between smolt survival for each pair of populations and d is the pairwise distance between each pair of populations at the point of marine entry. The parameter ν is the e-folding scale, or distance at which correlations are expected to decrease by 37% (e^{-1}), and ρ_0 is the expected correlation for populations with $d = 0$. The e-folding scale, although an arbitrary measure, has been used by other researchers, and thus provides a direct comparison of the scale of spatial coherence among species and studies (Pyper, Mueter, Peterman, Blackbourn, & Wood, 2002; Kilduff, Botsford, & Teo, 2014; Zimmerman et al., 2015). Distances between population pairs were measured in a geographic information system (GIS) as the shortest distance within saltwater between points of marine entry. Model parameters were estimated for the Salish Sea and non-Salish Sea population pairs separately using non-linear least squares, with pairwise observations weighted per the number of years of available survival data. Only pairwise correlations between populations with a 15-year minimum temporal overlap in survival time series were included.

To further examine spatial coherence in survival patterns, we conducted a cluster analysis based on the estimated Euclidean distance between annual estimates of population-specific smolt survival using Ward's hierarchical clustering algorithm (Legendre & Legendre, 2012). We used the R package *pvclust* (Suzuki & Shimodaira, 2011) to provide the approximately unbiased support for each node in the dendrogram, expressed as the proportion of bootstrapped dendrograms ($N = 1{,}000$) containing each node. Nodes with approximately unbiased p-values >.95 were considered strongly supported.

We evaluated 10 potential survival groupings among the Chinook salmon populations included in our study based on geographic location (e.g., Salish Sea versus outside the Salish Sea), hatchery release strategy (subyearling versus yearling), and run timing (spring versus summer-fall) by fitting the general form of the MARSS model that allows for evaluating specific informed hypotheses about the predominant spatial structure governing variability in survival (Holmes, Ward, & Wills, 2013; Holmes, Ward, & Wills, 2012). Here, the vector of observed marine survival at time t ($\mathbf{y}_t$) is modeled such that

$$\mathbf{y}_t = \mathbf{Z}\mathbf{x}_t + \mathbf{D}\mathbf{d}_t + \mathbf{v}_t \quad \text{where } \mathbf{v}_t \sim \mathrm{MVN}(\mathbf{0}, \mathbf{R}) \quad (4)$$

where $\mathbf{Z}$ is an $n \times m$ matrix containing 1s and 0s to indicate whether or not a particular time series is an observation of a latent trend ($\mathbf{x}_t$), $\mathbf{D}$ contains the estimated effects of user specified covariates at time t ($\mathbf{d}_t$) described in more detail below, and $\mathbf{v}_t$ is a vector of

TABLE 2 List of all Chinook salmon populations included in the study. A time series of smolt survival was generated using coded-wire tag recovery data for each population, with a minimum time series length of 20 years. Populations are divided by region (SS: Salish Sea, Northern, Southern) and basin (SOG, Strait of Georgia; PS, Puget Sound; JDF, Strait of Juan de Fuca; SEAK, Southeast Alaska; NBC, Northern British Columbia; ColR, Columbia River). Population numbers align with map in Figure 1. Time series are reported in terms of ocean entry year (OEY) with the total number of years represented by each time series in parentheses. Run timing (spring, summer-fall) and release strategy (Y, yearling; SY, subyearling) are noted

Region/Basin	Sub-basin	Population	OEY	Run timing	Release strategy
SS/SOG	Northern SOG	1. Quinsam Fall (QUI)	1975–2008 (34)	summer/fall	SY
SS/SOG	Northern SOG	2. Puntledge Summer (PPS)	1976–2009 (33)	summer/fall	SY
SS/SOG	Central SOG	3. Big Qualicum Fall (BQR)	1974–2009 (36)	summer/fall	SY
SS/SOG	Central SOG	4. Cowichan Fall (COW)	1986–2009 (22)	summer/fall	SY
SS/SOG	Central SOG (Fraser River)	5. Harrison Fall (HAR)	1982–2009 (27)	summer/fall	SY
SS/SOG	Central SOG (Fraser River)	6. Chilliwack Fall (Harrison Stock) (CHI)	1982–2009 (28)	summer/fall	SY
SS/SOG	Central SOG (Fraser River)	7. Nicola Spring (NIC)	1987–2010 (24)	spring	Y
SS/SOG	Central SOG (Fraser River)	8. Lower Shuswap Summer (SHU)	1985–2009 (25)	summer/fall	SY
SS/SOG	Southern SOG	9. Samish Fall Fingerling (SAM)	1975–2008 (26)	summer/fall	SY
SS/PS	Whidbey Basin	10. Skagit Spring Yearling (SKS)	1983–2009 (23)	spring	Y
SS/PS	Whidbey Basin	11. Stillaguamish Summer Fingerling (STL)	1981–2008 (23)	summer/fall	SY
SS/PS	Central Sound	12. Grovers Creek Fall Fingerling (GRO)	1982–2008 (27)	summer/fall	SY
SS/PS	Central Sound	13. Green River Fall Fingerling (GRN)	1972–2008 (32)	summer/fall	SY
SS/PS	South Sound	14. Nisqually Fall Fingerling (NIS)	1980–2008 (29)	summer/fall	SY
SS/PS	South Sound	15. South Puget Sound Fall Yearling (SPY)	1980–2009 (24)	summer/fall	Y
SS/PS	Hood Canal	16. George Adams Fall Fingerling (GAD)	1973–2008 (30)	summer/fall	SY
SS/JDF		17. Hoko Fall Fingerling (HOK)	1986–2007 (21)	summer/fall	SY
Southern/Coastal		18. Robertson Cr Fall (RBT)	1974–2009 (36)	summer/fall	SY
Southern/Coastal		19. Sooes Fall Fingerling (SOO)	1986–2007 (21)	summer/fall	SY
Southern/Coastal		20. Queets Fall Fingerling (QUE)	1978–2007 (29)	summer/fall	SY
Southern/ColR		21. Columbia Lower River H (LRH)	1977–2008 (32)	summer/fall	SY
Southern/ColR		22. Cowlitz Fall Tule (CWF)	1978–2008 (31)	summer/fall	SY
Southern/ColR		23. Lewis River Wild (LRW)	1978–2008 (28)	summer/fall	SY
Southern/ColR		24. Willamette Spring (WSH)	1977–2008 (32)	spring	Y
Southern/ColR		25. Spring Creek Tule (SPR)	1973–2008 (36)	summer/fall	SY
Southern/ColR		26. Lyons FerryYearling (LYY)	1986–2009 (21)	summer/fall	Y
Southern/ColR		27. Hanford Wild (HAN)	1987–2008 (22)	summer/fall	SY
Southern/ColR		28. Upriver Brights (URB)	1976–2008 (33)	summer/fall	SY
Southern/ColR		29. Columbia Summer (SUM)	1976–2008 (28)	summer/fall	SY
Southern/Coastal		30. Salmon River (SRH)	1977–2007 (30)	summer/fall	SY
Southern/Coastal		31. Elk River (ELK)	1978–2008 (31)	summer/fall	SY
Northern/NBC		32. Atnarko Summer (ATN)	1987–2008 (20)	summer/fall	SY
Northern/NBC		33. Kitsumkalum Summer (KLM)	1980–2008 (28)	summer/fall	SY
Northern/SEAK		34. Unuk Spring (UNU)	1984–2009 (21)	Spring	Y
Northern/SEAK		35. Alaska Central Inside (ACI)	1978-2009 (32)	Spring	Y
Northern/SEAK		36. Taku Spring (TAK)	1977–2009 (24)	Spring	Y

observation errors distributed as a multivariate normal with mean $\mathbf{0}$ and a diagonal, unequal variance-covariance matrix $\boldsymbol{R}$ (i.e., each time series is distributed independently, but not identically). Hypotheses regarding spatial groupings were evaluated by changing the elements in $\boldsymbol{Z}$ (0 or 1), with the columns representing groupings and the rows representing populations.

The latent trends are assumed to follow an autoregressive process, such that

$$\boldsymbol{x}_t = \boldsymbol{B}\boldsymbol{x}_{t-1} + \boldsymbol{w}_t \quad \text{where } \mathbf{w}_t \sim \text{MVN}(\mathbf{0}, \mathbf{Q}) \tag{5}$$

where $\boldsymbol{x}_t$ is a vector containing the values of the m latent smolt survival trends at time t, $\boldsymbol{B}$ is a matrix wherein the values along the diagonal reflect the degree of mean reversion (i.e., how fast the state

reverts to the mean after some perturbation) and $\boldsymbol{Q}$ is the variance-covariance matrix of the process errors. $\boldsymbol{Q}$ was specified with different values on the diagonal and 0's on the off diagonals to model the assumption of independent process errors and different variances for each subpopulation or group being tested. When testing a priori grouping hypotheses, we initially did not include any covariates, thereby setting matrix $\boldsymbol{C}$ to 0, which removed the term from the equation.

To further evaluate regional survival trends, we implemented dynamic factor analysis which is an alternative form of the MARSS model (Zuur et al., 2003). DFA is a dimension reduction technique similar to principal component analysis (PCA), but it is designed specifically for time series data. The general idea is to model n time series as a function of m latent (unobserved) temporal trends, where $m \ll n$. Thus, instead of the vector of observed marine survival at time t ($\boldsymbol{y}_t$) from Equation 4 being a function of a constrained form of $\boldsymbol{Z}$, specific to a grouping hypothesis, it is modeled as a linear combination of latent trends ($\mathbf{x}_t$) that are represented by an unconstrained form of $\boldsymbol{Z}$. This form of MARSS also assumes that the latent trends in survival follow a random walk process, thereby modifying equation (5) above by setting both the $\boldsymbol{B}$ matrix and variance-covariance matrix of the process errors ($\boldsymbol{Q}$) equal to the identity matrix (a diagonal matrix with 1s on the diagonal). Based on the predominant groupings in survival identified in the prior MARSS analysis (see results for details), we fitted separate models to survival time series of populations originating from four geographic areas including Northern (Alaska and Northern B.C.), Southern (Coastal and Strait of Juan De Fuca), and the two sub-basins within the Salish Sea including Strait of Georgia, and Puget Sound. To test whether Salish Sea populations may respond differently to continental scale environmental forcing than their coastal counterparts, we included as a covariate to the model the annual seasonal average North Pacific Gyre Oscillation (NPGO), a large-scale ocean circulation pattern that has recently been linked to annual variability in survival of west coast Chinook and coho salmon populations (Kilduff et al., 2015). We evaluated the data support for a maximum of two trends, both with and without annual NPGO included as a covariate, resulting in a total of four candidate models evaluated for each region.

Data support for all MARSS models was evaluated using the Akaike Information Criterion corrected for finite sample sizes (AICc). In each case, the model with the lowest AICc (ΔAICc = 0) was selected as the best explanatory model although models with a ΔAICc < 2 were considered to be similarly competitive explanatory models (Burnham & Anderson, 2002).

3 | RESULTS

Salish Sea Chinook salmon populations exhibited higher inter-population variability in smolt survival (coefficient of variation across entire, untransformed dataset = 1.45) than non-Salish Sea populations (CV = 1.21). Overall, Salish Sea populations exhibited weaker spatial coherence in survival than coastal populations. Specifically, we observed a greater rate of decline and a reduction in pairwise correlation in survival across increasing distances for Salish Sea relative to coastal populations (Figure 2; Table 3). Closer inspection revealed a higher frequency of relatively strong correlations ($r > .5$) among Southern Coastal populations relative to among Salish Sea populations (Figure 2e, f).

Cluster analysis results provided some evidence for regional-scale differences in survival between Salish Sea and non-Salish Sea populations, but exceptions to the general pattern confirmed high variability, particularly for Strait of Georgia populations. Two basal clusters were identified, one consisting of predominantly Salish Sea populations (hereafter "Salish Sea cluster"), and the other consisting of predominantly coastal populations (hereafter "coastal cluster"). Each cluster contained populations from multiple geographic regions, and populations from a single geographic region were distributed amongst multiple clusters (Figure 3). Five Strait of Georgia populations were grouped together in the Salish Sea cluster, and four Strait of Georgia populations were widely dispersed throughout the coastal cluster. All but one Puget Sound population was confined within the Salish Sea cluster. Of the fifteen Strait of Juan de Fuca and Southern Coastal populations, and the five Northern Coastal populations, all but three populations (all Southern Coastal) grouped with the coastal cluster. No association between release strategy and survival trend was observed as yearling populations were broadly distributed amongst several different clusters rather than grouped together.

Direct testing of a priori grouping hypotheses produced a best explanatory model based on geographic regions (Table 4; Figure 4). Groupings based on release strategy or run timing were poorly supported. The best model supported three or four regional groupings in survival with Strait of Georgia populations grouping with Southern Coastal and Strait of Juan de Fuca in the three-group model, and grouping separately in the four-group model. For the four-group model, the following geographic groupings were identified including Northern Coastal (n = 5), Southern Coastal + Juan de Fuca (n = 15), Strait of Georgia (n = 9), and Puget Sound (n = 7). Regionally-grouped models were more strongly supported than models representing a geographically-invariant (i.e., single grouping) hypothesis (Table 4).

The four regions identified by the MARSS models shared a general declining trend from 1980 to the early 1990s (Figure 4, Figure S1). This trend was most pronounced in the Strait of Georgia and least pronounced in the Puget Sound, where it appeared to be caused by a few low survival years during ocean entry years 1988–1990 (Figure 4). After the early 1990s, populations in the Southern Coastal + Juan de Fuca grouping exhibited more cyclical variability and increased synchrony in survival. In contrast, populations in the Northern Coastal grouping exhibited a largely flat trend from the early 1990s to present; Puget Sound and Strait of Georgia populations gradually rebounded from the low survival in the early 1990s (Figure 4). Regarding the raw, untransformed estimates, Puget Sound survival remained consistently low across the entire time series (rarely exceeding 5%), whereas some populations within each of the other four regions experienced periods of higher survival (Figure S1).

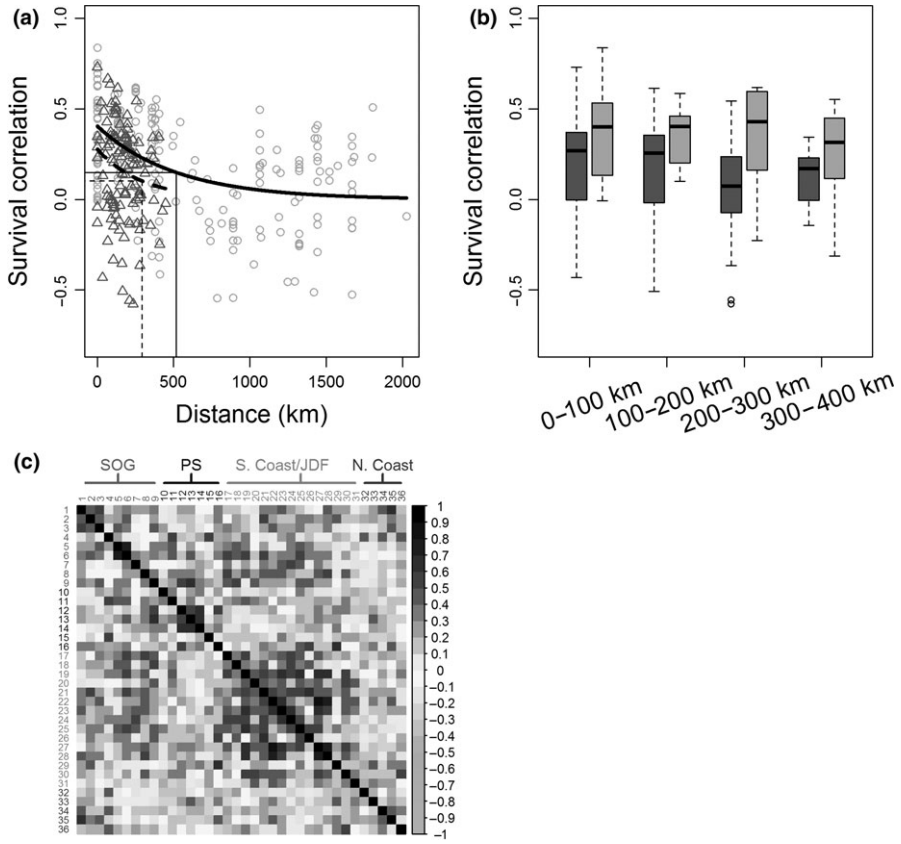


FIGURE 2 Regional covariation in survival for all 36 hatchery Chinook salmon populations for ocean entry years 1980–2008. (a) exponential decay models were fit to coastal populations (solid line, open light gray circles), and Salish Sea populations only (dashed line, open dark gray triangles) separately. The *e*-folding scale (ν) and the associated correlation predicted by each model are represented by the horizontal and vertical line segments. (b) boxplots comparing pairwise correlations in smolt survival Salish Sea (dark gray) and non-Salish Sea (light gray) populations across increasing distance between the point of marine entry of each population in 100-km increments up to 400 km. Here the thick line is the median, the boxes are the interquartile range, and the remainder of the data are contained in the whiskers and outlier points. The maximum pairwise distance for Salish Sea populations is approximately 400 km. (c) Pearson's pairwise correlations for all hatchery Chinook salmon populations organized by region including Strait of Georgia (1–9), Puget Sound (10–16), Strait of Juan de Fuca (17), Southern Coastal (18–31), and Northern Coastal (32–36). Number labels correspond to population information in Table 2 and geographic location in Figure 1. Survival data were logit-transformed and centered to mean = 0

TABLE 3 Parameter estimates from pairwise distance decay model fit to the relationship between Pearson's pairwise correlation in survival versus distance for Salish Sea populations and non-Salish Sea origin separately

	ν (*e*-folding scale)		ρ_0 (intercept)	
Period/Region	**Estimate**	**95% CI**	**Estimate**	**95% CI**
Salish Sea	292 km	151–1,916 km	0.28	0.16–0.41
Coastal	517 km	355–783 km	0.4	0.34–0.47

The prevalence of regional- and basin-scale asynchrony in Chinook salmon smolt survival was further supported by DFA. A single common trend was identified for the Northern Coastal, Southern Coastal, and Puget Sound areas, and two were identified for the Strait of Georgia (Figure 5; Table S1). The DFA trends for each region roughly matched the four survival trends estimated from these same regions produced by the MARSS models (Strait of Georgia: trend 1 only). Most populations exhibited positive loadings on each of the identified trends, indicating synchrony in the survival dynamics within each grouping, although two of the five populations within Northern Coastal Grouping and two of the seven populations within the Puget Sound grouping showed negative loadings. Of the four geographic groupings, only the top model for the Southern coastal grouping retained annual average NPGO as a driver of survival that accounted for additional unexplained variability (Table S1). The survival rates of populations within the Southern Coastal grouping were positively correlated with the annual NPGO index (Figure S2).

4 | DISCUSSION

We used multiple analytical approaches to investigate spatial and temporal coherence in the smolt survival of 36 Chinook salmon

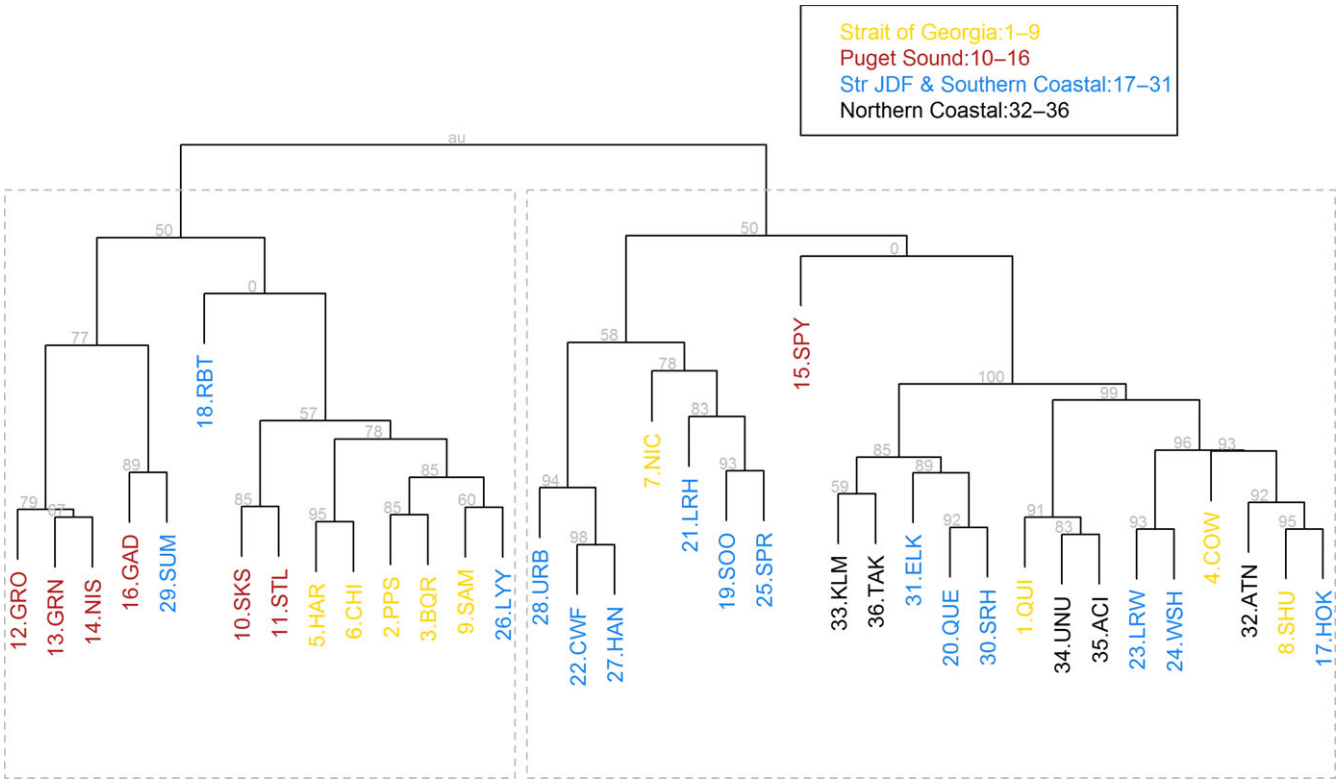


FIGURE 3 Dendrogram of temporal trends in smolt survival of 36 Chinook salmon populations. Survival data were logit-transformed and centered to mean = 0 prior to cluster analysis. On the vertical axis, line height represents the magnitude of difference between pairs of populations. Bootstrap support for each cluster is provided as an approximately unbiased *p*-value (significance at $p \geq .95$)

populations. Each approach provided evidence for greater spatial coherence in smolt survival among populations at the region/basin scale than at the continental scale. Importantly, the degree of spatial coherence was substantially weaker for Salish Sea populations compared to non-Salish Sea populations. This effect was because of the high degree of inter-population variability in survival observed among Salish Sea Chinook salmon (Figures 2 and 3). MARSS models testing different hypotheses on the predominant explanatory scale of variability in smolt survival provided evidence for spatial coherence at the regional/basin scale (Table 4; Figure 4). Regional differences in shared trends identified from DFA models suggest that coherence in annual survival rates is lower among regions. In general, estimated trends show that survival of Chinook salmon declined from 1980 through the early 1990s and increased moderately through 2008 (Figures 4 and 5). For the duration of this period, only in a single year (1987) did average Puget Sound survival approach 5%, whereas populations in the other three geographic areas frequently approached or exceeded 5% (Figure S1). Furthermore, we found regional differences in the effect of the NPGO index on annual survival rates, with a strong support for a positive association between the NPGO index and the survival dynamics of Southern Coastal populations only, suggesting that local factors may be more important in regulating survival of populations within the Salish Sea and Northern areas (Table S1 and Figure S2). Although we sought to identify the overarching spatial scale across which Chinook salmon populations exhibit covariation in survival, our analysis provided evidence that sub-basin, basin, and regional scales must be considered simultaneously; Kilduff et al. (2015) concluded that continental scale processes also influence survival.

Research evaluating spatial and temporal scales of survival covariation in other salmon species provides informative comparisons to our results. In general, our analysis of Chinook salmon provided considerably less evidence for fine-scale spatial coherence in smolt survival than a parallel investigation of Salish Sea coho salmon reporting stronger geographic association in survival (Zimmerman et al., 2015). Specifically, Zimmerman et al. (2015) reported a much higher correlation among proximate populations in the distance-decay model ($\rho_0 = 0.84$ across entire time series) than our analysis of Chinook salmon (Salish Sea = 0.28, Coastal = 0.40). In contrast, Pyper et al. (2002) identified a degree of synchrony among individual chum salmon populations ($\rho_0 = 0.44$) similar to what we estimated for Chinook salmon.

Differences among species may be due in part to differences in the degree of life history diversity exhibited by each species (Quinn, 2005). Coho salmon primarily enter the marine environment at age-1 in this region, and mature predominantly at age three after spending only 18 months in the ocean. In contrast, Chinook and Chum salmon enter the marine environment at age-0 (excepting the few yearling Chinook salmon stocks we analyzed), and exhibit multiple ages at maturity and variable ocean rearing lengths. Furthermore, coho salmon spend less time rearing in estuarine or nearshore environments compared to subyearling Chinook and chum salmon. Perhaps the

TABLE 4 Model selection results for multivariate autoregressive state space (MARSS) models testing for one to five hypothesized groupings (m = number groups) based on common trends of temporal variability in smolt survival among 36 Chinook salmon populations from within and outside the Salish Sea (t_{min} = 20 years). Models are shown in order of increasing ΔAICc relative to the top ranked model (ΔAICc = 0) and cumulative AICc weights

Model	Parameters	Groups	ΔAICc	Cumulative AICc Weight
4 groups (SEAK & NBC-SOG-PS- Southern & JDF)	45	4	0.00	0.50
3 groups (SEAK & NBC-Southern & JDF & SOG-PS	43	3	0.12	0.97
3 groups (SEAK & NBC-SOG & PS-Southern & JDF)	43	3	6.53	0.99
4 groups (SEAK & NBC-SOG-PS & JDF-Southern)	45	4	7.79	1.00
2 groups (SEAK & NBC-Southern & JDF & SOG & PS	41	2	16.14	1.00
3 groups (SEAK & NBC-Southern & PS & JDF-SOG)	43	3	23.05	1.00
2 groups (spring-summer/fall)	41	2	26.14	1.00
2 groups (subyearling – yearling)	41	2	29.47	1.00
2 groups (PS & SOG – JDF & Southern & SEAK & NBC)	41	2	32.09	1.00
1 group	39	1	35.25	1.00

younger age, and therefore smaller size, at which Chinook and chum salmon enter marine waters, combined with their greater use of nearshore and estuarine habitats, subject them to a greater degree of local influence on smolt survival compared to coho salmon.

In the present study, temporal covariation in early ocean survival for Chinook salmon was much more strongly influenced by geographic region or basin than by release strategy or run timing. Although uncertainty about the total number of geographic groupings remains, a geographically-invariant survival trend was poorly supported compared to models tested with region and basin scale groupings, suggesting that Salish Sea populations respond to basin-scale environmental variability. Several different hypotheses could explain this finding. Notably, many Chinook salmon populations reside within 100–200 km of their source river systems until their second year at sea (Orsi & Jaenicke, 1996; Trudel et al., 2009; Tucker et al., 2011, 2012), so common rearing habitat may persist for a year or more. Therefore, populations from the same basin likely share common rearing habitat within the first few months of marine entry and may be affected by more localized environmental factors (Ohlberger et al., 2016; Hertz et al., 2016). Regional specificity in the foraging ecology of juvenile Chinook salmon supports this hypothesis (Hertz et al., 2015).

Our results suggest that population scale processes are likely more pronounced for Salish Sea origin populations compared to coastal populations. Although we did not directly evaluate the influence of more localized environmental drivers in our study, only survival rates for Southern Coastal populations (and not Salish Sea populations) were linked to the continental-scale NPGO. Interestingly, Kilduff et al. (2015) identified a strong positive correlation between the NPGO and survival of hatchery Chinook salmon populations from the eastern Pacific Rim from central California to southeast Alaska, including Salish Sea populations. One possible explanation for this discrepancy is that Kilduff et al. (2015) utilized regional average annual survival rates for their analyses whereas we included survival rates for individual populations. Although region- and basin-scale marine conditions such as sea surface temperature can be driven by larger climatic processes occurring at the scale of the North Pacific Ocean (Moore et al., 2008b), which in turn can affect the survival rates of Chinook salmon (Hertz et al., 2016; Sharma et al., 2013), their effects may be dampened by localized environmental drivers within the Salish Sea.

We speculate that the synergistic effects of habitat loss (Good et al., 2005), long-term increases in predator abundance (Chasco et al., 2017), and poor water quality (Meador, 2014) contributed to the lack of covariation in survival among Salish Sea Chinook salmon populations. Magnusson and Hilborn (2003) observed higher early marine survival of coastal Oregon hatchery Chinook salmon in more pristine estuaries. As a result of the greater human population density and patterns in land use (e.g., agriculture and nearshore armoring), the quantity and quality of Salish Sea estuarine habitats and their marine subsidies (material recruitment, production of nearshore prey) are likely more variable than coastal systems outside the Salish Sea, which may contribute to higher inter-population variability in survival (see Figure 2). For example, within the Puget Sound, the estuary of the Green/Duwamish River Basin is wholly within the urbanized boundary of Seattle whereas the estuary of the Skagit River, although modified from its historic condition, is considerably more intact (NMFS, 2006). Furthermore, pelagic ecosystem changes within the Puget Sound have been associated with high levels of human development pressures not present in many coastal systems included in our study (Greene et al., 2015).

In the Salish Sea, outmigrant sub yearling Chinook smolts encountering poor habitat conditions may move offshore before achieving the body size or growth trajectory that maximizes survival potential. Duffy and Beauchamp (2011) demonstrated that the body size of juvenile Chinook salmon captured offshore in Puget Sound during July was a strong predictor of total marine survival, emphasizing the importance of early marine growth. Fish experiencing poor growth conditions in the estuary or nearshore may become more

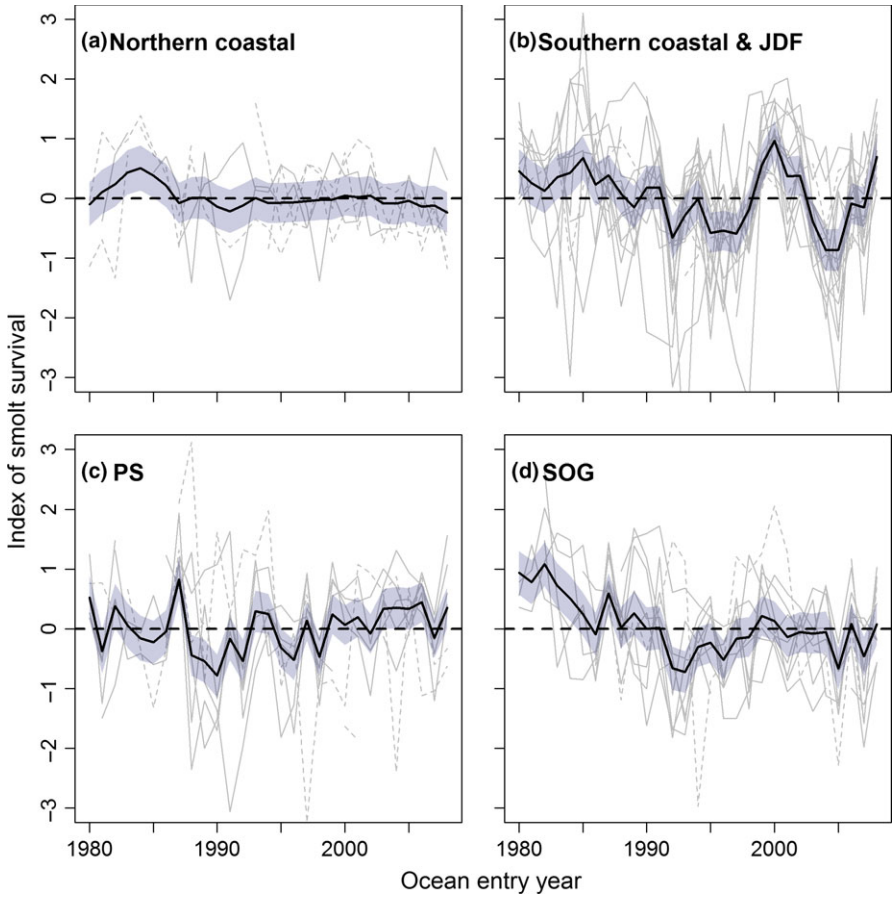


FIGURE 4 Fitted smolt survival values (black lines) with 95% confidence intervals (shaded area) estimated by the best-fit multivariate autoregressive state space (MARSS) model identifying four distinct groupings among 36 hatchery Chinook salmon populations according to the following geographic areas (a) Northern Coastal (SEAK/NBC), (b) Juan de Fuca & Southern Coastal (WCVI/WA/OR Coast), (c) Puget Sound, and (d) Strait of Georgia. Solid (subyearling) and dashed (yearling) gray lines are observed normalized survival values. [Colour figure can be viewed at wileyonlinelibrary.com]

vulnerable to predation because of increased time and energy spent foraging. The abundance of harbor seals within Puget Sound increased substantially in recent decades, resulting in a corresponding increase in total annual consumption of Chinook salmon smolts, from 1.0 million in 1970 to 8.5 million in 2015 (Chasco et al., 2017). Owing to the variation in the quality and quantity of estuarine habitat as well as patchy predator distributions, the effects of seal predation on long-term survival may not be uniform throughout the Salish Sea, further contributing to asynchrony in survival between geographically proximate populations within the Salish Sea.

Migratory pathways and residence times influence early marine survival patterns (Furey, Vincent, Hinch, & Welch, 2015; Melnychuk, Welch, & Walters, 2010). Melnychuk et al. (2010) described significant variation among populations in early marine migration patterns, so any differences in migratory behavior between coastal and the Salish Sea populations might be a source for regional-scale survival variation. Notably, of the 14 southern coastal Chinook salmon populations included in our study, 9 populations were from the Columbia River basin (Table 2). Despite having the same point of marine entry, some of these populations exhibited differing temporal patterns of survival (Figure 3), suggesting that additional factors may affect the overall survival of individual populations such as in-river survival or variation in ocean migratory pathways (see Jorgensen, Ward, Scheuerell, & Zabel, 2016).

Other factors we did not address in our analyses may also impact smolt survival. For example, genetic factors may influence the performance of salmon populations (Unwin, Kinnison, Boustead, & Quinn, 2003; Braun, Moore, Candy, & Bailey, 2016). Unfortunately, our study and similar studies (Kilduff et al., 2014, 2015; Sharma et al., 2013) were unable to effectively account for the effects of hatchery breeding, rearing, and release practices, which are known to influence smolt survival (Satterthwaite et al., 2014). Furthermore, significant reductions in domestic fisheries targeting both Puget Sound and Strait of Georgia populations throughout the same period in our analysis may have affected the accuracy of smolt survival estimates of these populations. Specifically, in cases where the catch component of CWT recoveries has been significantly reduced, estimates of population-specific smolt survival relies on large expansions of adult escapement from a small sub-sample of CWT recoveries. This issue is most pronounced in low abundance populations. For Puget Sound populations, high observation errors associated with the coded wire tag derived survival estimates may have contributed to our inability to detect any signal of covariation with other populations throughout the Salish Sea. Of the four geographic regions, survival rates for Puget Sound populations contained the highest average observation error estimated by the MARSS model (0.90 versus 0.76 for South coastal + JDF, 0.65 for SOG, and 0.39 for Northern populations).

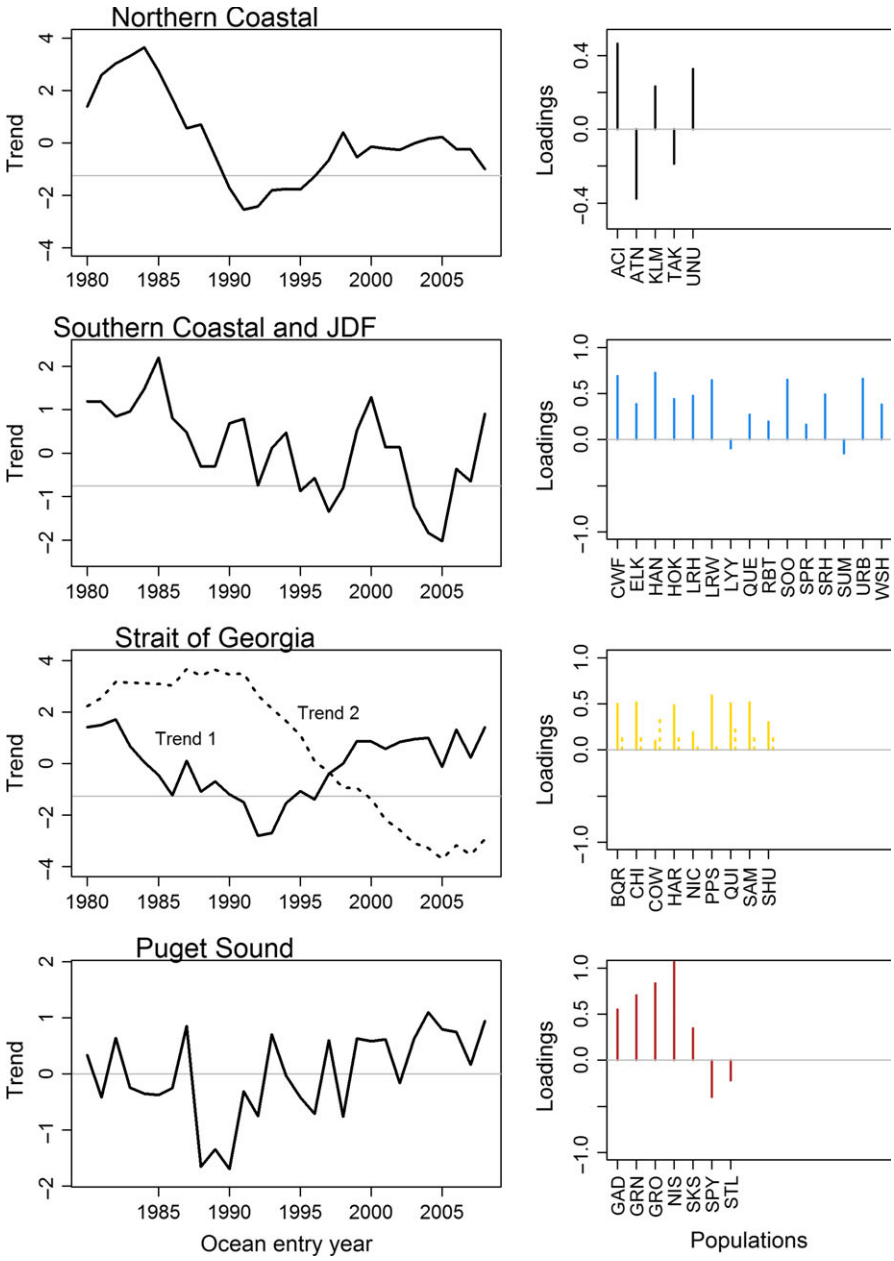


FIGURE 5 Estimated trends and factor loadings for individual DFA models fit separately to Northern Coastal (n = 5), Southern Coastal – JDF (n = 14), Strait of Georgia (n = 9), and Puget Sound (n = 5) population groupings

The high degree of inter-population variability in smolt survival of Chinook salmon, particularly among Salish Sea populations, presents challenges to fisheries managers. Harvest regulations in mixed-stock fisheries aim to minimize impacts on weaker populations while maintaining harvestable opportunity targeting more robust populations. Salish Sea populations make far-reaching northward ocean migrations and are vulnerable in mixed-stock commercial net, troll, and sports fisheries occurring from the West Coast of Vancouver Island to Southeast Alaska (PSC, 2015; Weitkamp, 2010). This underscores the importance of developing a cohesive monitoring framework that identifies physical and biological indicators of survival acting across multiple spatial and temporal scales relevant to the early rearing and marine life history of Chinook salmon. Such information will help inform robust management strategies aimed at protecting weak or threatened populations (e.g., Schindler & Hilborn, 2015). If managers focus on physical environmental or biological factors that affect survival at a single spatial scale, they will likely ignore variability caused by localized factors (Ohlberger et al., 2016). This would result in inaccurate abundance forecasts, thereby increasing the risk of either overexploitation of a population or foregone opportunity in specific fisheries.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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